Design Review One: Elastic Bandgap

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As the field of Materials Science grows even more complex, a thorough understanding of phonons is a necessary tool to pursue the development of new sound blocking materials. There are many applications for these potential elastic filters from blocking the sound of a crying baby to amplifying certain frequencies of sound. This study attempts to design a model that can select a material that can block sound of desirable frequencies. For the sake of simplicity, a 2D diatomic lattice system was used to model an elastic filter, typically a 3D crystal. Hamiltonian systems were used to create a simulation that determines what sound frequencies will be blocked. This "band gap" depends on certain parameters associated with the structure of a material. The implications of our model will be discussed in the following paper. It is concluded that certain sound frequencies cannot be blocked by real materials on the periodic table.

I. INTRODUCTION:

Controlling how sound travels through our living, working, and leisure spaces is of great interest. From absorbing sound in a practice room, to diffusing sound throughout a room for better acoustics, sound filters are widely used. In this model Hamiltonian equations were used to describe the atomic motions of a system and understand what frequencies could be blocked. Vibrations travelling through a medium have temporal and spatial characteristics that can be fully described with dispersion relations; this is temporal oscillation as a function of spatial periodicity for different vibrational modes of a system. In this system two ω values were calculated using Hamiltonian equations. The ω values derived determined the optical and acoustic frequencies. The values of these two terms were used to determine the band gap of the material, which is calculated by determining the smallest distance between the omega surfaces at the edge of the Brillouin zone. The size of the band gap determines the range of phonon frequencies at which the system will not oscillate. This means that the sound waves of that frequency will not be transmitted through the material.

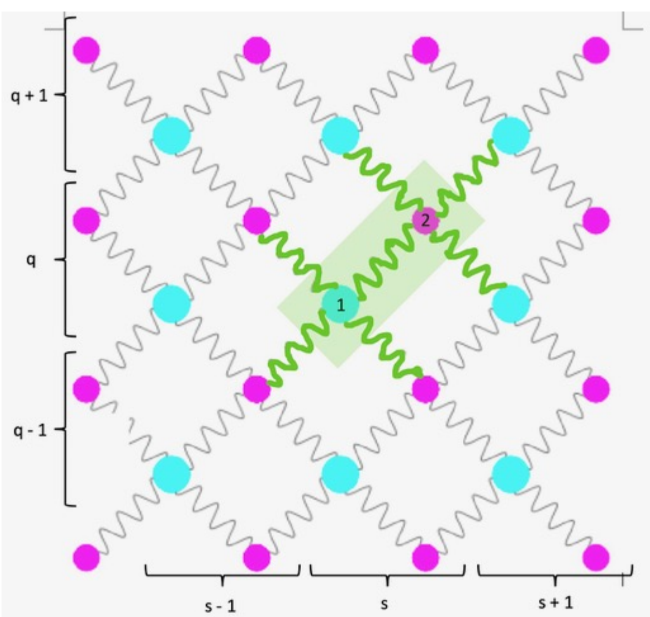

FIG. 1. Unit cell of model employed

II. METHODS:

To simplify our problem, a model of a 2D elastic filter was constructed despite the fact that most materials are 3 dimensional. The bonds were modeled as springs with spring constant "k", and the atoms were assumed to be point masses; this further simplified the work. The example that was taken into consideration was an orthorhombic lattice with two types of masses connected as seen in Figure 1. Hamiltonian kinetics were used to tackle the calculations. Hamiltonian mechanics requires expression for the kinetic energy of the masses, and potential energy of the springs. The springs highlighted in green represent the bonds that were included in our unit cell. The atoms were indexed using q, and s as shown in Figure 1. Below are the expressions that were found for the kinetic energy and potential energy.

$$\mathrm{KE} = \frac{1}{2}\left(\frac{P_{1q}^2}{m_1} + \frac{P_{1s}^2}{m_1} + \frac{P_{2q}^2}{m_2} + \frac{P_{2s}^2}{m_2}\right) \tag{1}$$

$$\begin{aligned} V_x = &\frac{1}{2}k\Big((X_{2(s,q)} - X_{1(s,q)})^2 + (X_{1(s+1,q-1)} - X_{2(s,q)})^2 \\ &+ (X_{1(s,q-1)} - X_{2(s,q)})^2 + (X_{2(s-1,q)} - X_{1(s,q)})^2 \\ &+ (X_{1(s+1,q)} - X_{2(s,q)})^2 + (X_{2(s,q+1)} - X_{1(s,q)})^2 \\ &+ (X_{2(s-1,q+1)} - X_{1(s,q)})^2\Big) \end{aligned} \tag{2}$$

$$V_y = \frac{1}{2}k\Big((Y_{2(s,q)} - Y_{1(s,q)})^2 + (Y_{1(s+1,q-1)} - Y_{2(s,q)})^2 + (Y_{1(s,q-1)} - Y_{2(s,q)})^2 + (Y_{2(s-1,q)} - Y_{1(s,q)})^2 + (Y_{1(s+1,q)} - Y_{2(s,q)})^2 + (Y_{2(s,q+1)} - Y_{1(s,q)})^2 + (Y_{2(s-1,q+1)} - Y_{1(s,q)})^2 \Big) \tag{3}$$

The Hamiltonian is merely the sum of these kinetic energy and potential energy terms. Once the Hamiltonian is found, the following rules can be applied:

$$H = \sum_i KE_i + \sum_i V_{X\,i} + \sum_i V_{Y\,i} \tag{4}$$

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$$\begin{aligned} \frac{du_{1(s,q)}}{dt} &= \frac{\partial H}{\partial p_{1(s,q)}}, & \frac{du_{2(s,q)}}{dt} &= \frac{\partial H}{\partial p_{2(s,q)}} \\ \frac{dp_{1(s,q)}}{dt} &= -\frac{\partial H}{\partial u_{1(s,q)}}, & \frac{dp_{2(s,q)}}{dt} &= -\frac{\partial H}{\partial u_{2(s,q)}} \end{aligned} \tag{5}$$

Some algebra yields four differential equations which can then be represented by a matrix.

$$\alpha = \frac{-4k}{m_1} + \omega^2 \tag{6}$$

$$\beta = \frac{-4k}{m_2} + \omega^2 \tag{7}$$

$$\gamma = \frac{k * \left(1 + e^{-ia_1k_x} + e^{ia_2k_y} + e^{-ia_1k_x + ia_2k_y}\right)}{m_1} \tag{8}$$

$$\varepsilon = \frac{k * \left(1 + e^{ia_1k_x} + e^{-ia_2k_y} + e^{ia_1k_x - ia_2k_y}\right)}{m_2} \tag{9}$$

$$\begin{bmatrix} \alpha & \gamma & 0 & 0 \\ \varepsilon & \beta & 0 & 0 \\ 0 & 0 & \alpha & \gamma \\ 0 & 0 & \varepsilon & \beta \end{bmatrix} \begin{bmatrix} X_1 \\ X_2 \\ Y_1 \\ Y_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} \tag{10}$$

To find the dispersion relation ω_k, the determinant of this matrix is set equal to zero and solved. Since we have 4 equations and 4 unknowns, this yields four expressions for the angular frequency ω, two of which are purely negative. These are neglected because they are simply a reflection of the positive ω expressions and the notion of a negative frequency is not valuable. After obtaining the dispersion relation, realistic values for elements on the periodic table can be incorporated. A carbon atom weighs about $2 * 10^{-26}kg$ and a lead atom weighs $3.4 * 10^{-25}kg$. These were the extrema of the masses considered. Common spring constant values range between 200N/m and 300N/m; 300N/m was the value used throughout this study.

III. RESULTS

Incorporating realistic mass, lattice constant and k values into the model yielded a certain range of values for ω. These values were far larger than 5000Hz. No parameters were able to yield a band gap even close to 20,000Hz (the highest frequency audible to human ears). Additionally the speed that sound would travel through the material was calculated using equation 11.

This model can also be used inversely to approximate the band gap of a known material. For example, this model indicates that a material made of Niobium (Nb) and Oxygen (O) could block phonons with frequencies between about 14 and 34 THz.

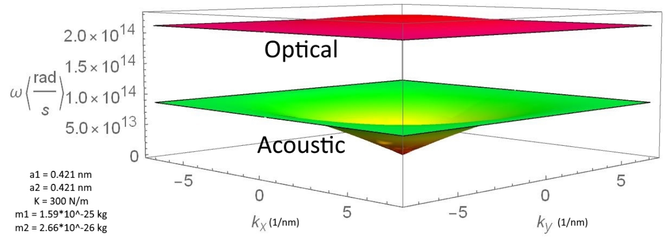

FIG. 2. Plot of dispersion relation for different k_x and k_y values

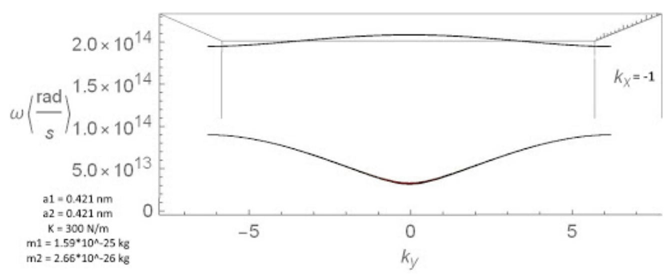

FIG. 3. 2D slice of dispersion plot at $k_x = -1$

$$V_g = \lim_{k_X, k_Y \to 0} \frac{d\omega_{acoustic}}{dk_{vector}} \tag{11}$$

Lattice constants (a1,a2)	2 to 15 nm	Range of common lattice constants
Atomic masses (m1,m2)	$2 * 10^{-26}$ to $4.5 * 10^{-25} Kg$	Range of atomic masses for elements
Spring Constant	200 to 300 N/m	Strength of typical bonds

These values are all ultrasonic however and therefore can have no applications in blocking sound that is audible to humans. Neither is it possible to achieve an average speed of sound of 20000 - 60000 m/s through the material. Further research is required to develop models for more complex crystal systems represented by more complex spring and mass lattices in order to potentially find materials that can achieve phonon blocking in the audible range.

The high values found for the optical branch suggest that its name is a bit of a misnomer because light at those frequencies is actually invisible to the human eye. The lower branch however, is acoustic because these are more common ω values for sound and will most likely be occupied by sound waves.

IV. CONCLUSION/LIMITATIONS

Extensive exploration of the model developed in this study shows that there are no realistic materials that can block frequencies of sound within 150 THz and 250 THz. However, this model does make many assumptions. It is not necessarily the case that atoms can be modeled as point masses, and the bonds as ideal springs. The model is also treated as a 2 dimensional lattice, so definitive conclusions cannot necessarily be drawn about 3 dimensional materials. In the future, these calculations could be extended to 3 dimensions and the system can be analyzed again.

Because this model makes many assumptions, another more rigorous model could find a solution. For instance, it is not necessarily the case that atoms can be treated as point masses. In fact, unit cells are often assumed to be made up of rigid spheres to calculate the atomic packing factor. Modelling bonds as ideal springs is also not particularly convenient because "spring constant" cannot be determined easily for any material in the way atomic mass can. This is likely only a valid approximation at small perturbations of the lattice, so perhaps this model is more reliable for modelling materials at lower temperatures. Most importantly, because this model looked at a 2 dimensional lattice, definitive conclusions regarding 3 dimensional materials cannot be drawn. In the future, these calculations could be extended to 3 dimensions and the system could be reanalyzed. More exploration could be done to assess the validity of this model by comparing to laboratory experiments.

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